

1.2 Measurements & Errors

Question Paper

Course	CIEA Level Physics
Section	1. Physical Quantities & Units
Topic	1.2 Measurements & Errors
Difficulty	Easy

Time allowed: 50

Score: /35

Percentage: /100

Question 1a

A physics teacher tells a class:

"Measurements may give a precise value for the quantity being determined but this may not necessarily be an accurate value"

Describe what is meant by

(i)
accuracy

[1]

(ii)
precision

[1]

[2 marks]

Question 1b

Draw lines on Fig. 1.1 to show how accuracy and precision are affected by different types of error



Fig. 1.1

[3 marks]

Question 1c

The teacher uses the diagram shown in Fig. 1.2 to illustrate the difference between accuracy and precision.

The circles represent targets A, B, C and D and the dots represent arrows hitting the targets.

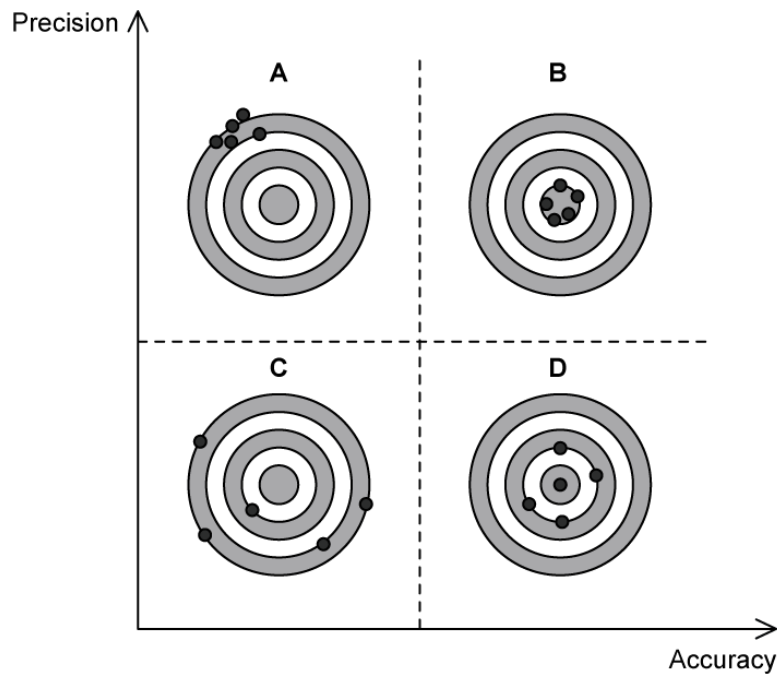


Fig. 1.2

Complete the following sentences to explain how targets A, B, C and D represent differing degrees of accuracy and precision.

neither accurate nor precise accurate but not precise
precise but not accurate both accurate and precise

A is
B is
C is
D is

[4 marks]

Question 1d

The temperature of the air in a room is measured using a mercury-in-glass thermometer.

For each of the following, tick [✓] one box to indicate whether the experimental technique would improve accuracy, precision or neither. The first row has been completed as an example.

	improves accuracy	improves precision	improves neither
correcting for an incorrectly calibrated thermometer	✓		
taking repeat readings of temperature and finding an average			
using a thermometer with a linear scale			
keeping your eye in line with the scale and the liquid level for each temperature reading			

[3 marks]

Question 2a

Distinguish between random and systematic errors.

[2 marks]

Question 2b

From a set of metal discs, one has a mass of 9.5 g and a thickness of approximately 2 mm.

State the instrument that should be used to measure

- (i)
the mass of one or more discs [1]
- (ii)
the thickness of one disc [1]
- (iii)
the thickness of a stack of 20 discs. [1]

[3 marks]

Question 2c

A student takes five repeat readings of the thickness of the disc. The readings are shown in Table 1.1.

Table 1.1

thickness / mm	2.012	2.001	1.998	2.008	1.994
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Using the readings in Table 1.1, calculate

- (i)
the mean value of thickness [2]
- (ii)
the range of the results [1]
- (iii)
the uncertainty in the mean value of thickness. [2]

[5 marks]

Question 2d

State how the instrument used to measure thickness can

(i)
reduce or avoid a systematic error

[1]

(i)
reduce random errors

[2]

[3 marks]**Question 3a**

A student uses a stopwatch to measure the time taken for one complete swing of a pendulum.

They measure the time for 10 complete oscillations to be 16.4 ± 0.1 s.

Calculate

(i)
the mean value of the time taken for one complete swing

[1]

(ii)
the absolute uncertainty in this mean value

[1]

(iii)
the percentage uncertainty in this mean value.

[2]

[4 marks]

Question 3b

The student is using the pendulum to determine a value for the acceleration of free fall g .

As well as the period T of oscillation, they also take measurements of the length L of the pendulum.

The value obtained, with its uncertainty, is $L = 67 \pm 1$ cm.

Calculate the percentage uncertainty in the measurement of the length L .

[2 marks]**Question 3c**

The relationship between T , L and g is given by

$$g = 4\pi^2 \frac{L}{T^2}$$

Using your answers in **(a)** and **(b)**, show that the percentage uncertainty in the value of g is 2.7%.

[2 marks]

Question 3d

There are some occasions where the resolution of an instrument is not the only limiting factor of uncertainty in a measurement.

State

- (i)
the meaning of the 'resolution of an instrument'
- (ii)
one limiting factor that affects the uncertainty in the measurement of time using a stopwatch.

[1]

[1]

[2 marks]

Model Answers: Easy

1a

(a)

(i) Accuracy is defined as a measure of:

- How close a measured / calculated value is to the true value; [1 mark]

(ii) Precision is defined as a measure of:

- How close the measured values are to one another **OR** the consistency of values obtained by repeated measurements; [1 mark]

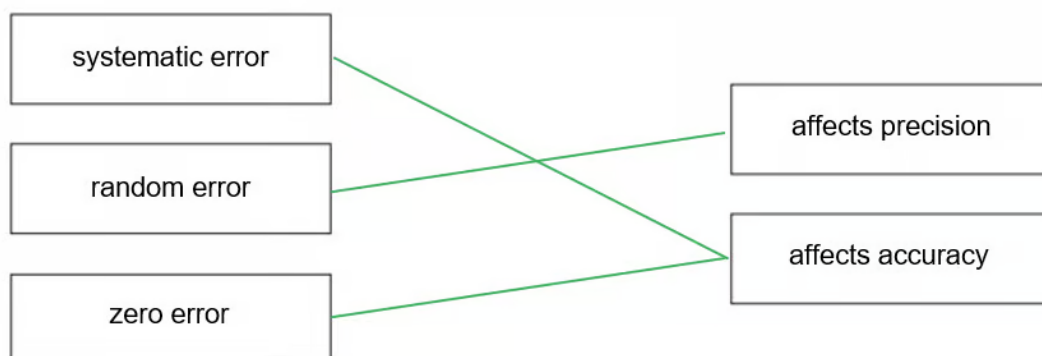
[Total: 2 marks]

Make sure not to confuse **precision** with **resolution** i.e. how many significant figures the measurement would be given to is to do with resolution, not precision

1b

(b) The correct answers are:

One mark for each correct line:



[Total: 3 marks]

1c

(c) The correct sentences are as follows:

- A is **precise but not accurate**; [1 mark]
- B is **both accurate and precise**; [1 mark]
- C is **neither accurate nor precise**; [1 mark]
- D is **accurate but not precise**; [1 mark]

[Total: 4 marks]

High precision means a small spread of values

This is shown by the dots located close together i.e. targets **A** and **B**

Low precision means a large spread of values

This is shown by the dots located far apart from one another i.e. **C** and **D**

High accuracy means close to the true value

This is shown by the dots located near the centre i.e. targets **B** and **D**

Low accuracy means not close to the true value

This is shown by the dots located away from the centre i.e. **A** and **C**

1d

(d) The correct table is as follows:

	improves accuracy	improves precision	improves neither
correcting for an incorrectly calibrated thermometer	✓		
taking repeat readings of temperature and finding an average		✓	
using a thermometer with a linear scale			✓

keeping your eye in line with the scale and the liquid level for each temperature reading	✓		
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- *One mark for each correct tick*

[Total: 3 marks]

Remember: **precise** readings are ones that are close **together**:

- Repeat readings can give very similar measurements and can make identifying anomalous results much easier
- Reading the scale at eye level reduces parallax error and ensures that all readings are taken in the same way

Remember: **accurate** results are ones that are close to the **true value**

- An incorrectly calibrated thermometer would make all the readings higher or lower by the same amount, hence it would make the final value higher or lower by a set amount

2a

(a) Distinguish between random and systematic errors:

Systematic errors

Any **one** from:

- Cause all readings to differ from the true value by the same amount; [1 mark]
- Cannot be eliminated by averaging; [1 mark]
- Caused by errors / faults in measuring instrument / experimental set-up; [1 mark]

Random errors

Any **one** from:

- Cause readings to spread / scatter around the true value; [1 mark]
- Can be eliminated by averaging (only if not stated for systematic error); [1 mark]
- Caused by factors which vary in an unpredictable way **OR** caused by errors due to experimenter / observer; [1 mark]

[Total: 2 marks]

For random errors the most common wrong answer is 'human error'. This is a common phrase which we all understand, so it's tempting to say it here.

However, in our physics experiments, it could be argued that **all** errors are due to human error. If the experimenter forgot to check their measuring scales were at zero before starting, the error introduced to the data is systematic, but the mistake was made by a human!

To avoid this, **never** say 'human error' in your physics exam. Be more specific about the nature of the problem.

2b

(b)

(i) The best instrument for measuring mass is:

- A digital / top-pan balance; [1 mark]

(ii) The best instrument for measuring the thickness of 1 disc is:

- Micrometer / screw gauge / digital callipers; [1 mark]

(iii) The best instrument for measuring the thickness of 20 discs is:

- Vernier callipers; [1 mark]

[Total: 3 marks]

It's useful to have an idea of the maximum possible readings of equipment as well as their lowest i.e. the resolution, so that you can make the best judgement for the most appropriate equipment to use in different scenarios

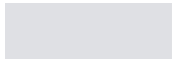
The line between when to use a micrometer or vernier callipers can get blurred, a good range to use for each is

- Micrometer: 0.01 mm to 2.5 cm
- Vernier callipers: 0.1 mm to 15 cm

For lengths between 10 – 30 cm, a 30 cm ruler is usually recommended, and anything above that should be measured using a metre rule

2c

(~)



(ii) The best instrument for measuring the thickness of 1 disc is:

- Micrometer / screw gauge / digital callipers; [1 mark]

(iii) The best instrument for measuring the thickness of 20 discs is:

- Vernier callipers; [1 mark]

[Total: 3 marks]

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For lengths between 10 – 30 cm, a 30 cm ruler is usually recommended, and anything above that should be measured using a metre rule

2c

(c)

(i) The mean value of thickness is:

- Mean = the sum of all the readings divided by the number of readings

- Mean = $\frac{2.012 + 2.001 + 1.998 + 2.008 + 1.994}{5}$ [1 mark]

- Mean = 2.0026 = 2.003 mm [1 mark]

(ii) The range of the results is:

- Range = the difference between the highest and lowest value
- Range = 2.012 – 1.994 = 0.018 mm [1 mark]

(iii) The uncertainty in the mean value of thickness is:

- Uncertainty in mean value = $\pm$ half the range
- Uncertainty in mean value = $\pm (\frac{1}{2} \times 0.018)$ [1 mark]
- Uncertainty in mean value = ± 0.009 mm [1 mark]

[Total: 5 marks]

Remember the mean value should always be quoted to the same number of **significant figures** as the readings

2d

(d) The instrument used to measure thickness is the micrometer:

(i) Reduce or avoid a systematic error by:

- Checking for a zero error / zeroing the micrometer before use; [1 mark]

(ii) Reduce random errors by:

- Taking (at least) three readings; [1 mark]
- At different points around the disc; [1 mark]

[Total: 3 marks]

3a

(a)

(i) Calculate the mean time for one complete swing:

- Time of 1 oscillation = $\frac{16.4}{10} = 1.64$ s [1 mark]

(ii) Determine the absolute uncertainty:

- The absolute uncertainty must also be divided by 10
- Absolute uncertainty = $\pm \frac{0.1}{10} = \pm 0.01$ s [1 mark]

(iii) Calculate the percentage uncertainty:

- % uncertainty = $\frac{\text{uncertainty}}{\text{value}} \times 100\%$
- % uncertainty = $\frac{0.01}{1.64} \times 100\%$ [1 mark]
- % uncertainty = $0.6098 = 0.61\%$ [1 mark]

$\therefore$ mean time period $T = 1.64 \pm 0.01 \text{ s}$ or $1.64 \text{ s} \pm 0.61\%$ (2 s.f.)

[Total: 4 marks]

The number of significant figures you quote calculated uncertainties to is very important

A significant figure is a reliably **known** digit, whereas an uncertainty is an **estimate**. Therefore, the percentage uncertainty should be given to the **same** number of significant figures as the result

And remember, uncertainties should **never** be given to **more** significant figures than the result as it **cannot** be known to a greater degree of precision than the result itself

3b

(b) Calculate the percentage uncertainty in the measurement of L :

- % uncertainty = $\frac{\text{uncertainty}}{\text{value}} \times 100\%$
- % uncertainty = $\frac{1}{67} \times 100\%$ [1 mark]
- % uncertainty = $1.4925 = 1.5\%$ [1 mark]

$\therefore$ length $L = 67 \pm 1 \text{ cm}$ or $67 \text{ s} \pm 1.5\%$ (2 s.f.)

[Total: 2 marks]

3c

(c) Show that the percentage uncertainty in the value of g is 2.7%:

List the known quantities:

- % uncertainty in $L = 1.5\%$
- % uncertainty in $T = 0.6\%$

Calculate the % uncertainty in g :

- % uncertainty in $g = \text{\% uncertainty in } L + (2 \times \text{\% uncertainty in } T)$ [1 mark]
- % uncertainty in $g = 1.5 + (2 \times 0.6)$ [1 mark]
- % uncertainty in $g = 2.7\%$

[Total: 2 marks]

Make sure you learn the rules for calculating uncertainties

In the relationship $g = 4\pi^2 \frac{L}{T^2}$ there is division and T is raised to a power, and the factor $4\pi^2$ can be ignored (since it is a ratio not a measurement), so the rules for calculating the uncertainty in g will involve:

- Adding the % uncertainties together – this applies when multiplying or dividing data
- Multiply the % uncertainty by the power – this only applies to quantities that are raised to a power

3d

(d)

(i) Resolution of an instrument is defined as:

- The smallest observable / recognisable change in a reading (on the instrument); [1 mark]

(ii) The limiting factor of a stopwatch is:

- The reaction time of the observer / experimenter; [1 mark]

[Total: 2 marks]

